AI Impact on Employment

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ABSTRACT

This report explores the multifaceted impact of Artificial Intelligence (AI) on employment, focusing on job displacement, creation, and transformation across various sectors. Employing a quantitative methodology, we conducted a survey using closed-ended questionnaires and analyzed the data with the Statistical Package for Social Sciences (SPSS). Our findings reveal a complex landscape where AI serves both as a disruptor and an enabler in the labor market. While AI automates routine tasks, leading to job displacement in certain areas, it also creates new opportunities, particularly in sectors requiring advanced technical skills. The impact varies across industries and demographics, with some groups more susceptible to disruption. These insights underscore the necessity for proactive strategies, including reskilling and upskilling programs, to mitigate risks and harness the opportunities presented by AI. By understanding these dynamics, stakeholders can develop informed policies and initiatives to support workforce adaptation in an evolving technological landscape.

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1. Introduction

The primary objective of this report is to analyze the influence of AI on employment patterns, identifying sectors and demographics most affected, and proposing recommendations for stakeholders to navigate the evolving job market.

AI technologies have rapidly advanced, leading to increased automation and the integration of intelligent systems in various industries. While AI offers efficiency and innovation, it also poses challenges related to job security and workforce adaptation. Understanding these dynamics is crucial for policymakers, businesses, and workers alike.

This report focuses on the impact of AI on employment within the United States, with particular attention to sectors such as technology, manufacturing, and services. It encompasses an analysis of existing literature, survey data, and statistical evaluations to provide a comprehensive overview.

2. Objectives

- To assess the extent of AI-induced job displacement and creation across different sectors.
- To identify demographic groups most vulnerable to AI-driven employment changes.
- To evaluate existing strategies for workforce adaptation and propose actionable recommendations.

3. Problem Statement

The integration of AI into the workplace has led to significant shifts in employment structures, raising concerns about job displacement, skill mismatches, and economic inequality. There is a pressing need to understand these changes to develop effective policies and training programs that ensure inclusive growth and workforce resilience.

4. Project Scope

The project encompasses a mixed-methods approach, combining quantitative surveys with statistical analysis to explore AI's impact on employment. The study targets professionals across various industries in the pakistan, aiming to capture a diverse range of experiences and perceptions related to AI integration in the workplace.

5. Methodology

The research employs closed-ended questionnaires distributed to a sample population across within our department in multiple programs (Software Engineering and Computer Science). The collected data is analyzed using the Statistical Package for Social Sciences (SPSS) to identify patterns and correlations between AI adoption and employment outcomes. This methodological approach ensures a rigorous and data-driven analysis of the subject matter.

6. Existing Work

Several global organizations and studies have explored how Artificial Intelligence (AI) is impacting employment. These findings help us understand the possible effects AI could have on countries like Pakistan. Some of the most notable works include:

6.1. OECD Report on AI in the Workplace

The Organization for Economic Co-operation and Development (OECD) highlights how AI is transforming the nature of work. According to the report, AI is changing job roles and the types of skills needed. Routine tasks are becoming automated, which means workers need to adapt by learning new digital and problem-solving skills. The report emphasizes the importance of workforce training programs to prepare employees for AI-driven environments.

6.2. McKinsey's Analysis of AI Adoption

McKinsey & Company's research shows that AI adoption varies widely across industries. For example, sectors like finance, healthcare, and manufacturing are using AI to improve efficiency and productivity. While AI can create new jobs in data analysis, software development, and machine learning, it also risks replacing certain types of jobs, especially those involving repetitive tasks. McKinsey suggests that countries should invest in reskilling programs to prepare their labor force for this shift.

6.3. ILO's Global Study on AI and Jobs

The International Labour Organization (ILO) provides a global perspective on how AI affects both the number and quality of jobs. The study highlights that while AI may not cause mass unemployment, it will reshape work by changing job responsibilities and creating demand for new skills. It also notes that low- and middle-income countries may face bigger challenges in adapting to AI due to limited resources and digital infrastructure.

7. Findings

The analysis reveals that AI's impact on employment is nuanced, with certain sectors experiencing job displacement while others see job creation. Key findings include:

- **Job Displacement**: Routine and administrative roles are increasingly automated, leading to job losses in sectors such as manufacturing and clerical work.
- **Job Creation**: Emerging technologies have led to the creation of new roles, particularly in AI development, data analysis, and cybersecurity.
- **Skill Gaps**: There is a growing demand for digital literacy and advanced technical skills, highlighting the need for upskilling and reskilling initiatives.

8. Survey Results

The survey conducted as part of this study indicates that:

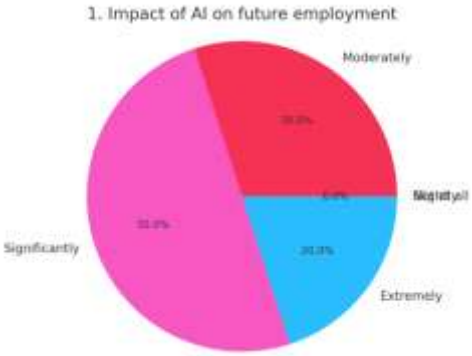

Figure 1: Impact of AI

- 50% believe significant impact of AI on future employment, while 30% favor moderately. However, 10% think it will touch extremely

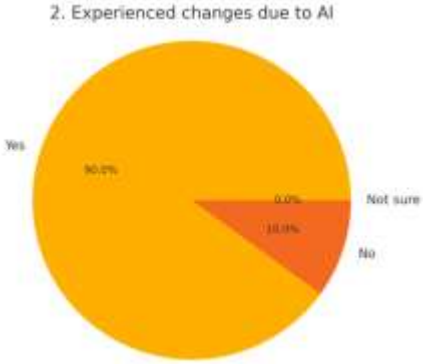

Figure 2: Experienced changes

- 90% of respondents have experienced changes in their jobs or studies due to the integration of AI technologies, indicating a significant and widespread impact.

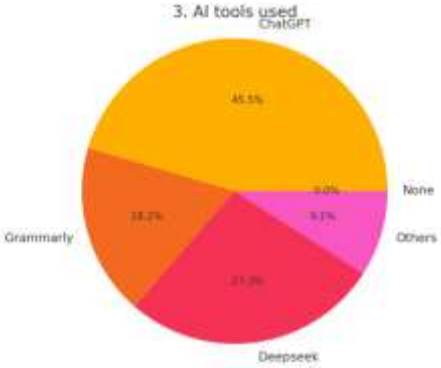

Figure 3: AI tools

- 80% have used at least one AI tool, with ChatGPT (50%) being the most used, followed by Deepseek (30%) and Grammarly (20%), highlighting growing AI adoption in academic and professional environments.

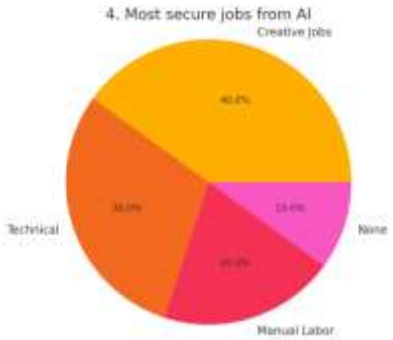

Figure 5: Most Secure Jobs

- 40% believe creative jobs are the most secure from AI disruption, while 30% favor technical roles. However, 10% think no jobs are truly safe, reflecting growing concerns over automation.

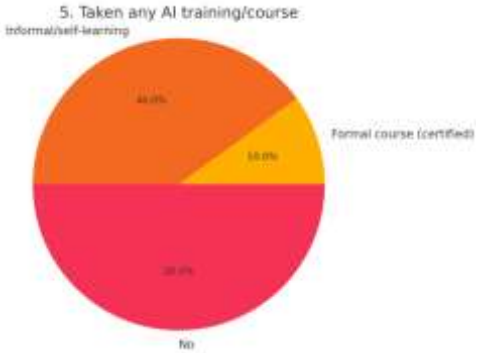

Figure 4: Taken any training

- When it comes to AI-related training, only 10% have taken certified courses, while 40% engaged in informal/self-learning. Still, 50% have not taken any training, suggesting a gap in preparedness.

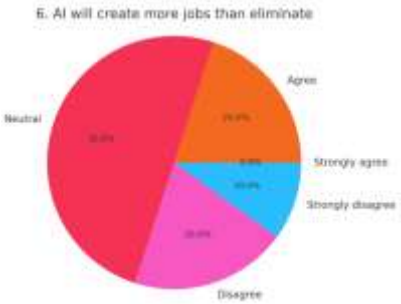

Figure 6: AI will create more jobs

- 70% remain neutral or uncertain about whether AI will create more jobs than it eliminates, with only 20% agreeing it will have a positive net impact. This reflects skepticism about AI's long-term employment benefits.

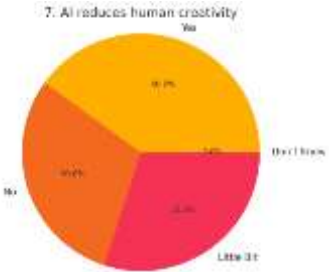

Figure 7: AI reduce creativity

- On creativity, 40% feel AI reduces human creativity at work, while another 30% believe it has minimal or no effect. This shows mixed perceptions about AI's role in creative processes.

Figure 8: Trust AI

- Trust in AI for hiring decisions is limited, with only 30% expressing trust, while 40% would trust it "a little bit" and 30% would not trust it at all.

Figure 9: Literacy of AI

- 80% agree or strongly agree that AI literacy is essential for future jobs, emphasizing the need for awareness and education to stay relevant in the job market.

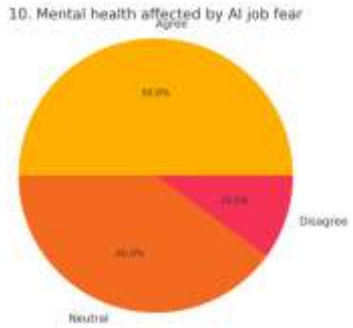

Figure 10: Mental health

- Half the respondents believe AI-related job insecurity impacts mental health, and 40% remain neutral. This suggests growing psychological concerns linked to AI integration in the workforce.

9. Recommendations

- **Policy Development**: Governments should formulate policies that support workforce transition, including funding for training programs and social safety nets.

- **Educational Reform**: Educational institutions must update curricula to include AI-related competencies, ensuring graduates are prepared for the modern workforce.
- **Corporate Responsibility**: Businesses should invest in employee development programs to facilitate internal mobility and skill enhancement.

10. Conclusion

AI's integration into the workplace presents both challenges and opportunities. While certain job roles are at risk of automation, new positions are emerging that require advanced skills. A collaborative approach involving policymakers, educators, and industry leaders is essential to navigate this transition effectively, ensuring that the workforce is equipped to thrive in an AI-driven economy.

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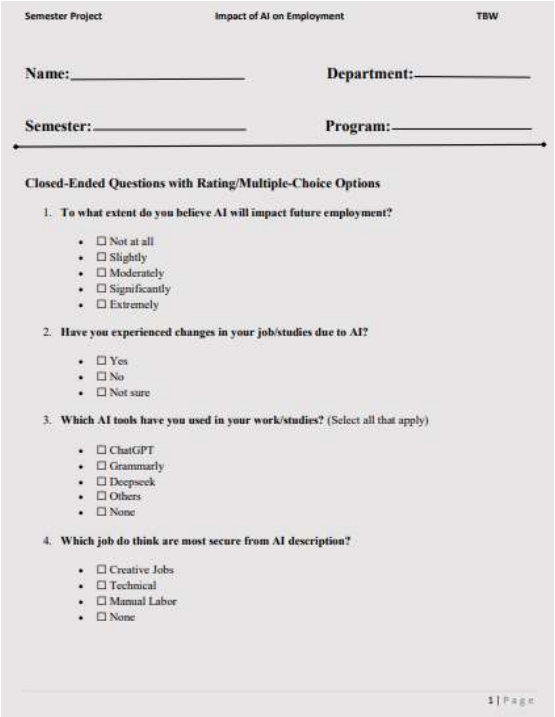
APPENDIX

Appendix A

Full Survey Questionnaire



Semester Project | Impact of AI on Employment | TBW

Name:_____ **Department:**_____

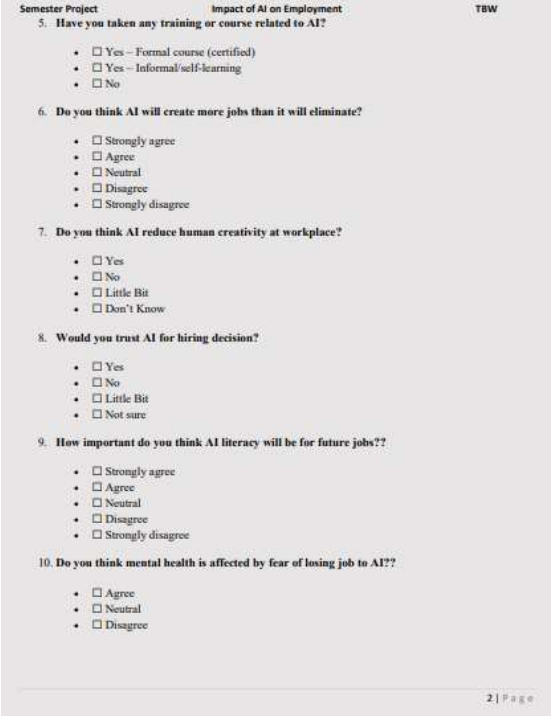
Semester:_____ **Program:**_____

Closed-Ended Questions with Rating/Multiple-Choice Options

1. **To what extent do you believe AI will impact future employment?**
   - ☐ Not at all
   - ☐ Slightly
   - ☐ Moderately
   - ☐ Significantly
   - ☐ Extremely
2. **Have you experienced changes in your job/studies due to AI?**
   - ☐ Yes
   - ☐ No
   - ☐ Not sure
3. **Which AI tools have you used in your work/studies?** (Select all that apply)
   - ☐ ChatGPT
   - ☐ Grammarly
   - ☐ Deepseek
   - ☐ Others
   - ☐ None
4. **Which job do think are most secure from AI description?**
   - ☐ Creative Jobs
   - ☐ Technical
   - ☐ Manual Labor
   - ☐ None

1 | Page

Figure 12: Questionnaire page 1



Semester Project | Impact of AI on Employment | TBW

5. **Have you taken any training or course related to AI?**
   - ☐ Yes – Formal course (certified)
   - ☐ Yes – Informal/self-learning
   - ☐ No
6. **Do you think AI will create more jobs than it will eliminate?**
   - ☐ Strongly agree
   - ☐ Agree
   - ☐ Neutral
   - ☐ Disagree
   - ☐ Strongly disagree
7. **Do you think AI reduce human creativity at workplace?**
   - ☐ Yes
   - ☐ No
   - ☐ Little Bit
   - ☐ Don't Know
8. **Would you trust AI for hiring decision?**
   - ☐ Yes
   - ☐ No
   - ☐ Little Bit
   - ☐ Not sure
9. **How important do you think AI literacy will be for future jobs??**
   - ☐ Strongly agree
   - ☐ Agree
   - ☐ Neutral
   - ☐ Disagree
   - ☐ Strongly disagree
10. **Do you think mental health is affected by fear of losing job to AI??**
    - ☐ Agree
    - ☐ Neutral
    - ☐ Disagree

2 | Page

Figure 11: Questionnaire page 2

Appendix B

Data Visualization (Chart outputs)

Appendix C

Raw Response Data (Statistical Package for Social Sciences)

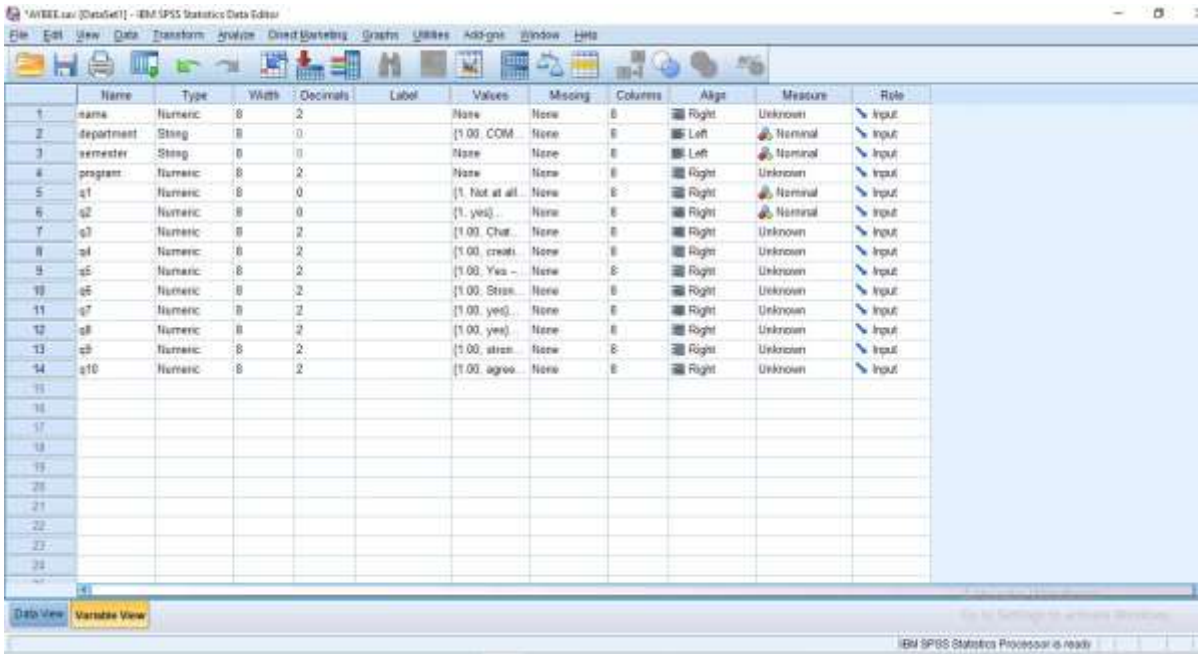


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Figure 13: SPSS tool